DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

UNIVERSITY OF BARISAL

Final EXAMINATION

Course Title: Simulation and Modeling

Course Code: CSE-3209

3rd Year 2nd Semester

Session: 2023-24 (Admission: 2021-22)

D, M, MT, V, EA

Time: 3 hour **Marks:** 60

Answer any five Questions from the followings.

1. a) What is computer simulation? Briefly describe the steps of simulation study. [4]

   b) Briefly explain the differences between discrete system and continuous system with suitable figure. [4]

   c) Write down the name of several entities, attributes, activities, events, and state variables for the following system i) A small appliance repair shop ii) A hospital emergency room [4]

2. a) Write short notes on the followings: [8]

   i) Exponential Distribution ii) Lognormal Distribution

   iii) Geometric Distribution iv) Bernoulli Distribution

   b) A Hurricane is to hit in the country, and expected to follow poisson distribution with a mean of 0.8 per year. Find the possibility of occurring more than two hurricanes in a year. Also find the possibility of exactly one hurricane in a year. [4]

3. a) Suppose that x and y are jointly discrete random variables with [3]

$$P(x,y)=(x+y)/30 \quad for\ x=0,1,2\ and\ y=0,1,2,3$$
$$=0 \qquad\qquad 0,\ therwise$$

Are x and y independent?

   b) Suppose that x and y are jointly continuous random variables with [6]

$$f(x,y)=y-x \quad \text{for } 0<x<1 \text{ and } 1<y<2$$
$$=0, \qquad otherwise$$

Compute E(x), Var(x), E(y), Var(y), Cov(x, y), Cor(x,y)

   c) Test for whether the 3rd, 8th, 13th, and so on, numbers in the following sequence at the beginning of this section are autocorrelated using $\alpha = 0.05$. [3]

0.12	0.01	0.23	0.28	0.89	0.31	0.64	0.28	0.83	0.93
0.99	0.15	0.33	0.35	0.91	0.41	0.60	0.27	0.75	0.88
0.68	0.49	0.05	0.43	0.95	0.58	0.19	0.36	0.69	0.87

[8]

4. a) Explain over aspects and characteristics of a Queueing System.

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b) Customers at a restaurant arrive in groups (one to eight persons). The number of persons (per group) for 300 customers and the relative frequencies appear are shown below. Draw empirical CDF. [4]

Arrivals per party	1	2	3	4	5	6	7	8
Frequency	30	110	45	71	12	13	7	12

5. a) Explain different methods of random number generation. [9]

b) What are the challenges in generating pseudo random numbers. [3]

6. a) What are the problems may occur during the generation of random numbers? [3]

b) Define linear congruential method for generating random number. Use the linear congruential method to generate a sequence of four (4) random numbers with X0 = 27, a = 17, c = 43, and m = 100. [4]

c) What are the methods used for testing a random number? Suppose, five numbers 0.44, 0.81, 0.14, 0.05, 0.93 are generated. Test the uniformity of the generated numbers using the Kolmogorov–Smirnov test with the level of significance $\alpha = 0.05$. [Critical value $D\alpha=0.565$] [5]

7. a) When Linear Congruential Generator is used? [2]

b) How Linear Congruential Generator is implemented? Explain with an example. [4]

c) Explain two methods of uniformity test with appropriate examples. [6]

8. At a grocery store with one counter, customers arrive at random from 1 to 8 minutes apart (each of inter-arrival time has the same probability of occurrence). The service times vary from 1 to 6 minutes with the probabilities as 0.15, 0.20, 0.30, 0.20, 0.10 and 0.05 respectively. Analyze the system by simulating the arrival and service of 15 customers. [Justifying your situation and requirements, you can choose your required random values] [12]

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